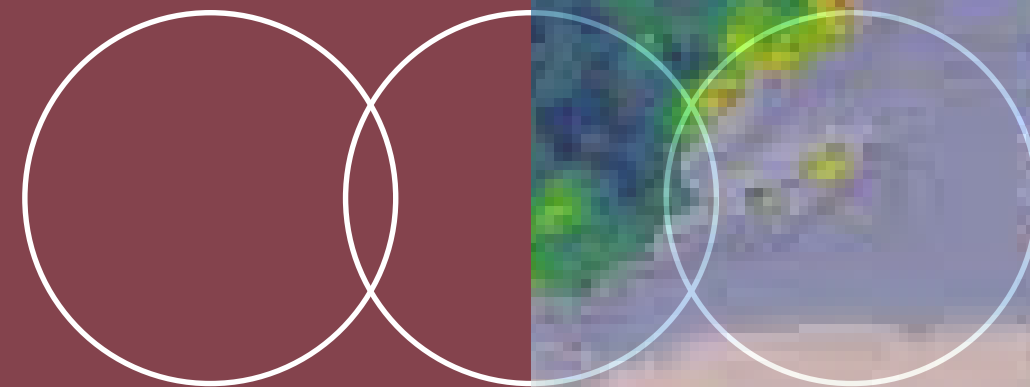


Analyzing Soil Pollution Worldwide

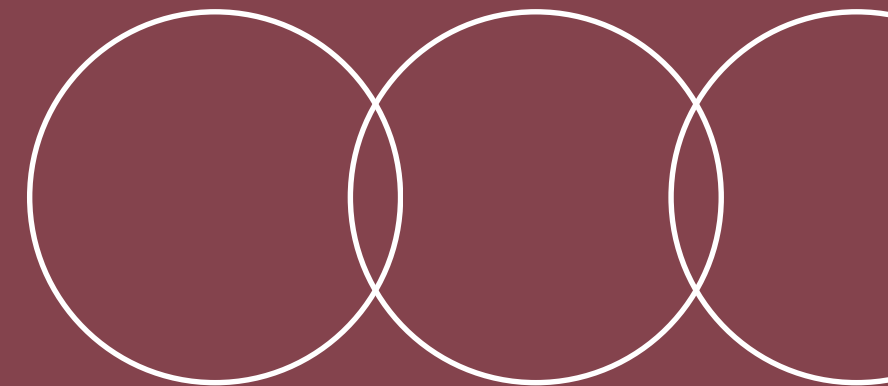
Presented by: **Mohamed Chandoul**
Badis Zammouri



About this Dataset

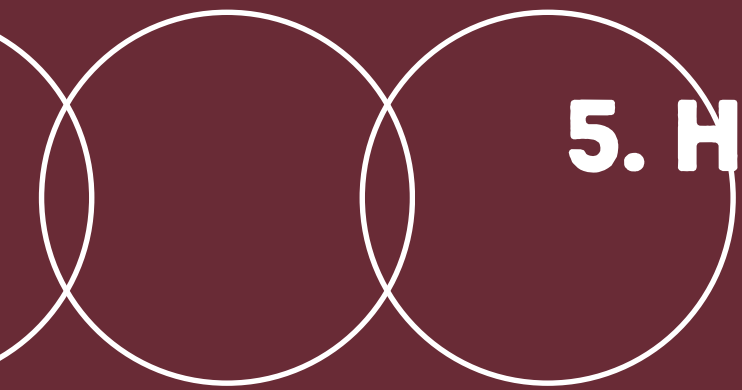
This dataset contains 3000 synthetic records simulating real-world scenarios of **soil pollution** and related **diseases**. It captures environmental, agricultural, and demographic variables to analyze correlations between soil contamination and human health outcomes.

The data was generated using **probabilistic models** and **domain knowledge**, making it suitable for exploratory analysis, machine learning, and environmental health research.



Challenge Outline

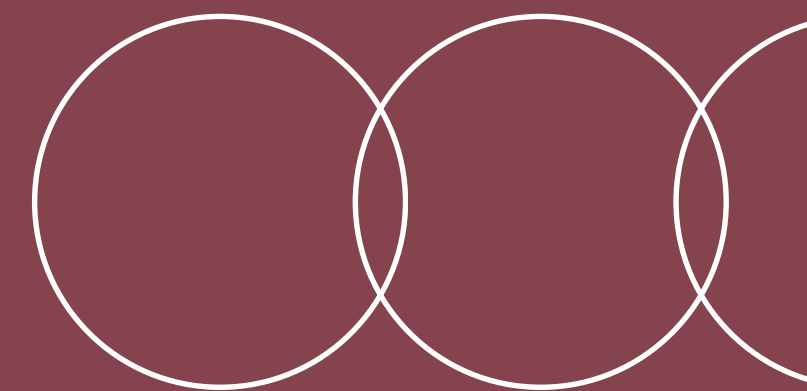
- 1. DATA CLEANING AND TRANSFORMATION.**
- 2. DATA VISUALIZATION.**
- 3. STATISTICAL ANALYSIS.**
- 4. USER FORMS FOR INPUTTING OR QUERYING DATA.**
- 5. HIGHLIGHT KEY INSIGHTS.**



1- a- Deleting useless columns and duplicates

```
Sub Remove_useless_columns()  
    Dim ws As Worksheet  
    Dim i As Long  
    Set ws = ThisWorkbook.Sheets("soil_pollution_diseases")  
  
    'Region: it does not match the real geography  
    'Crop_Type, Farming_Practice, Health_Symptoms, Soil_Texture: irrelevant to our analysis  
    ws.Columns("K").EntireColumn.Delete  
    For i = 1 To 19  
        If ws.Cells(1, i).Value = "Region" Or ws.Cells(1, i).Value = "Crop_Type" Or ws.Cells(1, i).Value = "Farming_Practice" Or ws.Cells(1, i).Value = "Health_Symptoms" Or ws.Cells(1, i).Value = "Soil_Texture" Then  
            ws.Columns(i).EntireColumn.Delete  
        End If  
    Next i  
End Sub
```

```
Sub Remove_duplicates()  
    Dim ws As Worksheet  
    Dim i As Long  
    Dim last_row As Long  
    Set ws = ThisWorkbook.Sheets("soil_pollution_diseases")  
    last_row = ws.Cells(Rows.Count, 1).End(xlUp).Row  
  
    For i = 3 To last_row  
        If ws.Cells(i, "C") = ws.Cells(i - 1, "C") And ws.Cells(i, "D") = ws.Cells(i - 1, "D") And ws.Cells(i, "E") = ws.Cells(i - 1, "E") Then  
            ws.Rows(i).EntireRow.Delete  
        End If  
    Next i  
End Sub
```



1-b- Handling missing values

```
Sub Handle_missing_values()  
    Dim ws As Worksheet  
    Dim last_row As Long  
    Dim last_column As Long  
    Dim i As Long  
    Dim j As Long  
    Dim k As Long  
    Dim n As Long  
  
    Set ws = ThisWorkbook.Sheets("soil_pollution_diseases")  
    last_row = ws.Cells(Rows.Count, 1).End(xlUp).Row  
    last_column = ws.Cells(1, Columns.Count).End(xlUp).Column  
  
    'since most of the values are not related to each other on the space and time (different dates/countries), using the mean of the previous and next values to fill missing ones will cover  
    For i = 3 To last_row  
        For j = 1 To last_column  
            If IsEmpty(ws.Cells(i, j)) Then  
                If IsNumeric(ws.Cells(1, j).Value) Then  
                    n = 0  
                    For k = 2 To last_row  
                        If ws.Cells(i, "C").Value = ws.Cells(k, "C").Value And ws.Cells(i, "D").Value = ws.Cells(k, "D").Value Then  
                            ws.Cells(i, j).Value = ws.Cells(i, j).Value + k  
                            n = n + 1  
                        End If  
                    Next k  
                    If n = 0 Then  
                        ws.Rows(i).EntireRow.Delete  
                    Else  
                        ws.Cells(i, j).Value = ws.Cells(i, j).Value / n  
                    End If  
                Else  
                    ws.Rows(i).EntireRow.Delete  
                End If  
            End If  
        Next j  
    Next i  
End Sub
```

1-c- Data sorting

```
Sub Sorting_Dates() '(Using Record Macro)
    Range("B2").Select
    ActiveCell.FormulaR1C1 = "9/12/2024"
    Range("B2").Select
    ActiveCell.FormulaR1C1 = "9/12/2024"
    Range("B2").Select
    ActiveCell.FormulaR1C1 = "9/12/2024"
    Range("B2:B3001").Select
    ActiveWorkbook.Worksheets("soil_pollution_diseases").Sort.SortFields.Clear
    ActiveWorkbook.Worksheets("soil_pollution_diseases").Sort.SortFields.Add Key _
        :=Range("B2"), SortOn:=xlSortOnValues, Order:=xlAscending, DataOption:= _
        xlSortNormal
    With ActiveWorkbook.Worksheets("soil_pollution_diseases").Sort
        .SetRange Range("A2:S3001")
        .Header = xlNo
        .MatchCase = False
        .Orientation = xlTopToBottom
        .SortMethod = xlPinYin
        .Apply
    End With
End Sub
```

2- Data Visualization

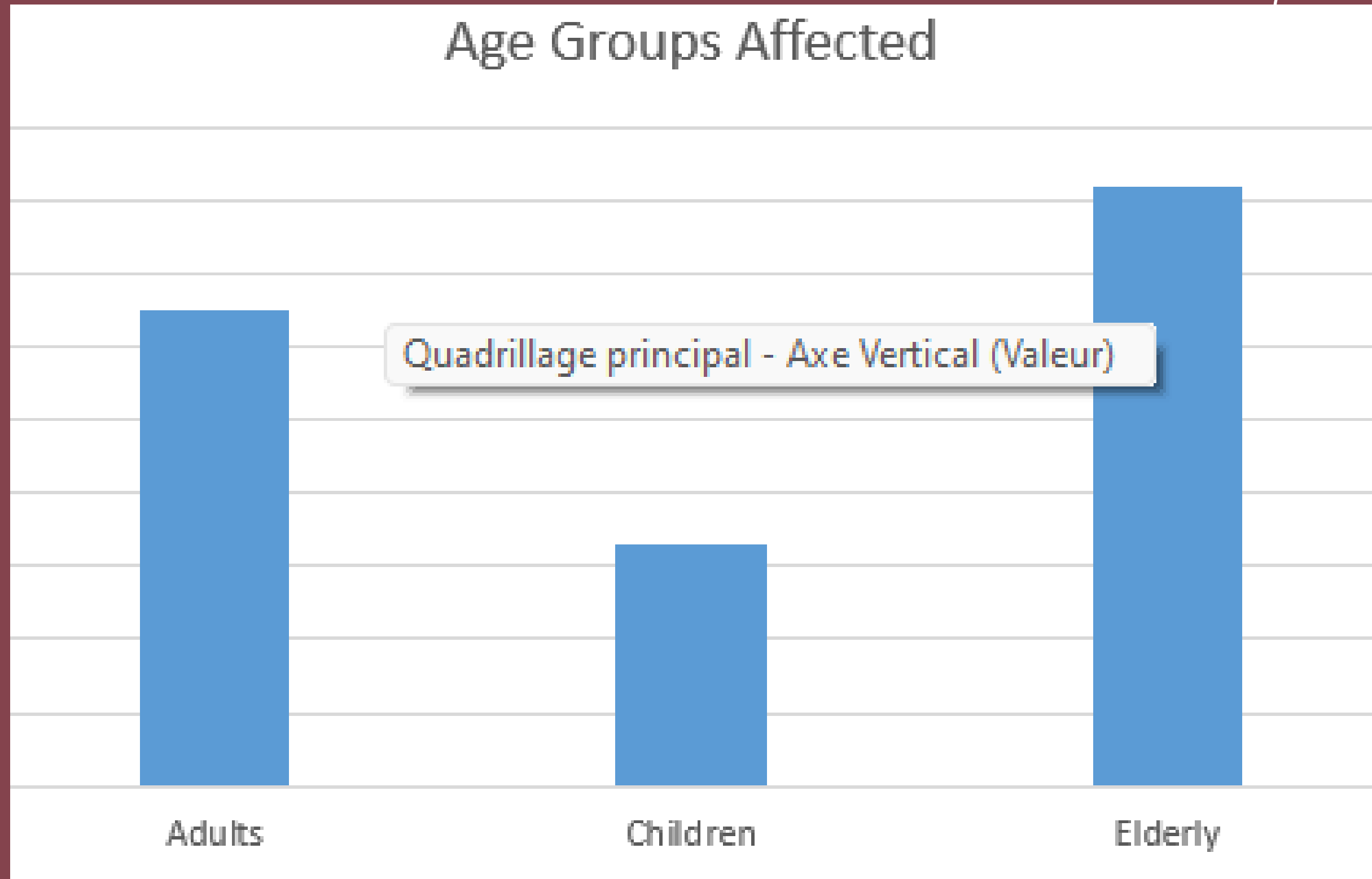


Chart 1: A comparaison between age groups affected by soil pollution.

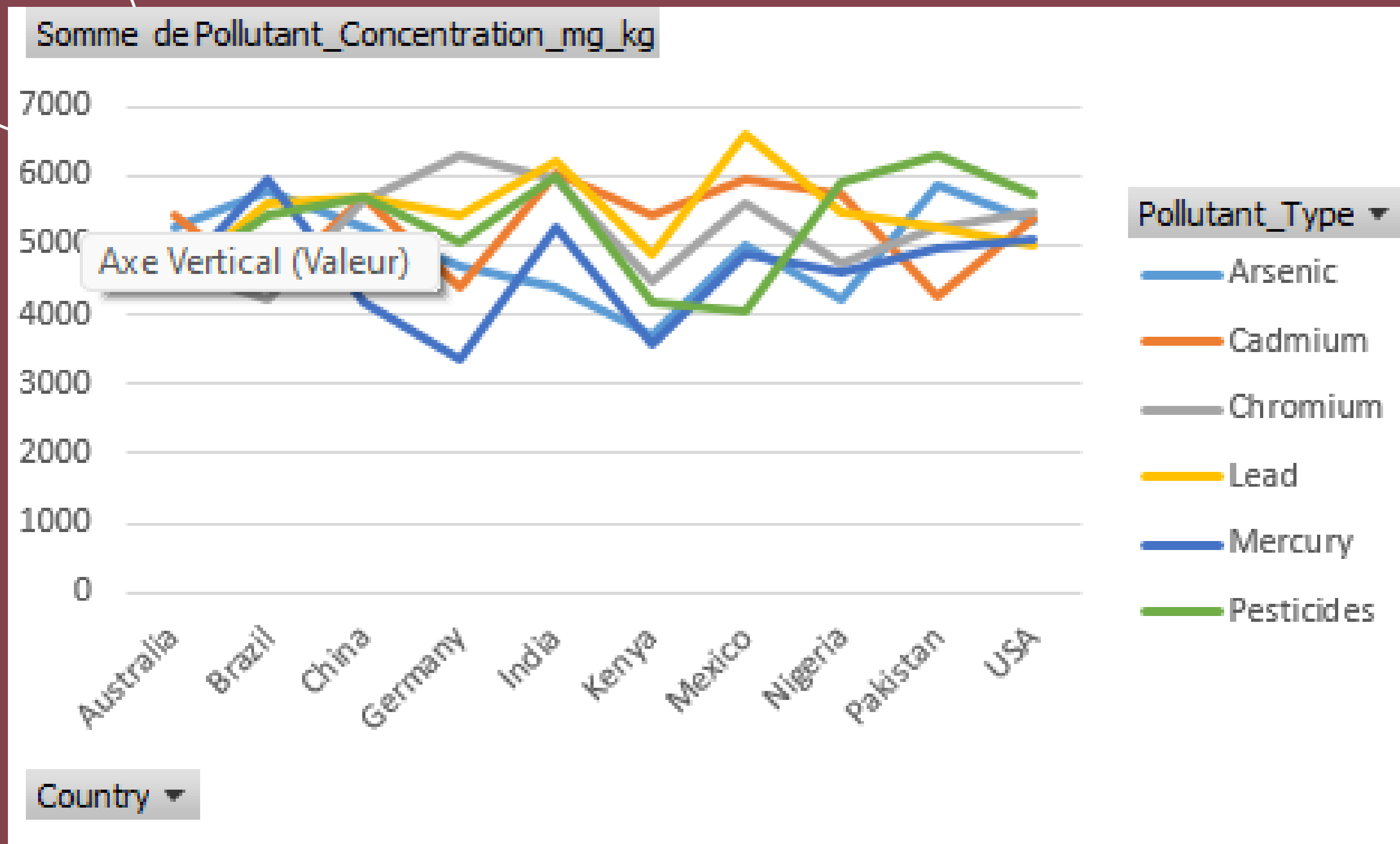
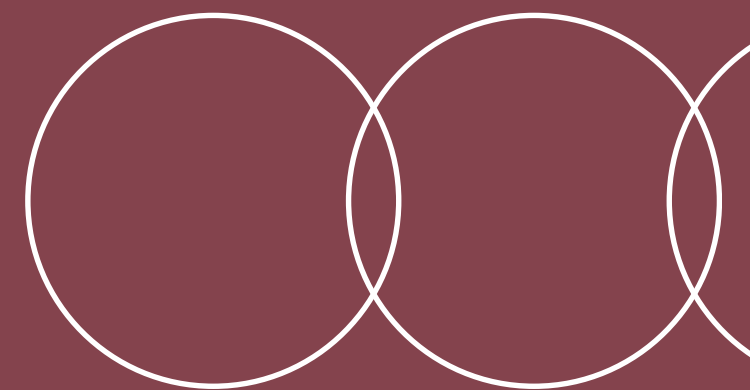


Chart 2: Countries and Polution Types.

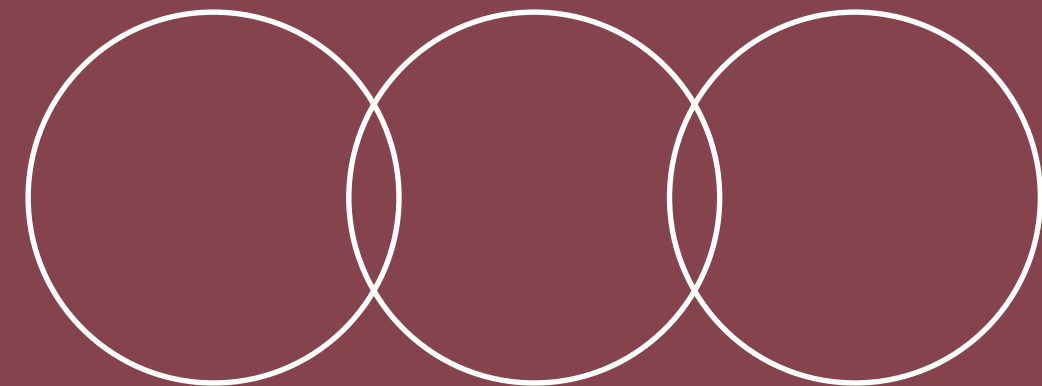
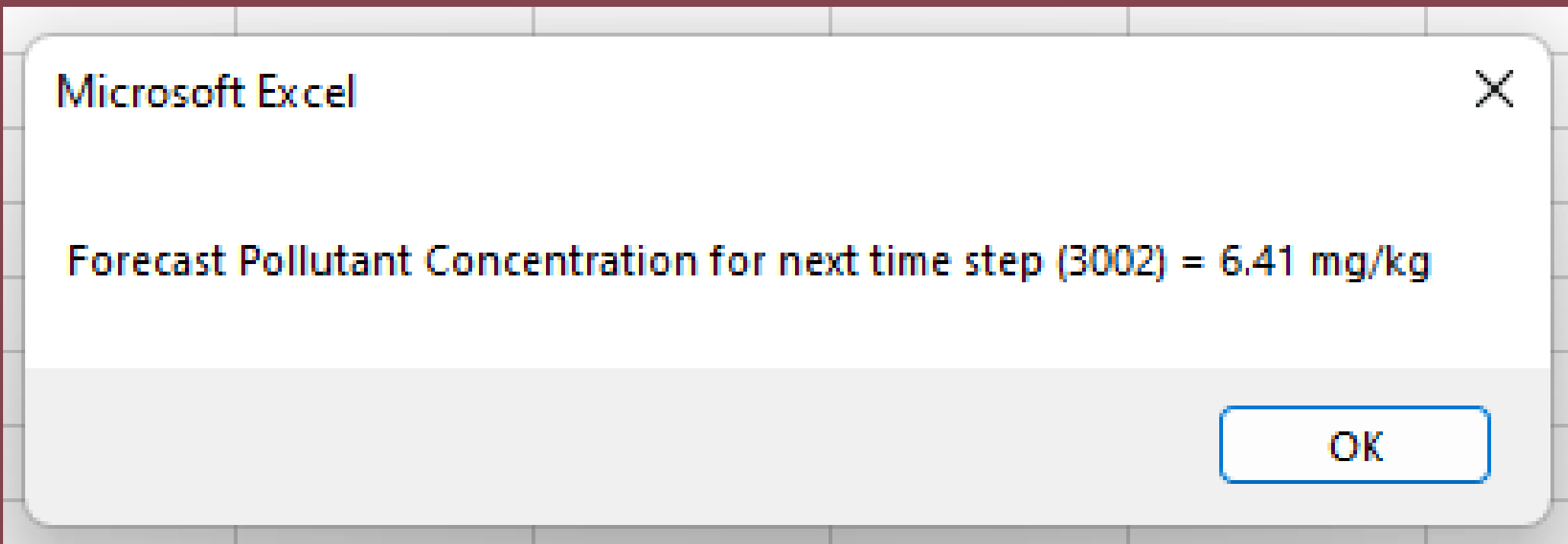
3-a- Calculating statistics

```
Private Sub CommandButton1_Click()  
    Dim ws As Worksheet  
    Dim i As Long  
  
    Set ws = ThisWorkbook.Sheets("soil_pollution_diseases")  
  
    m = WorksheetFunction.Average(Range("E2:E3001"))  
    MsgBox "Mean of pollutant Concentration = " & m  
End Sub
```



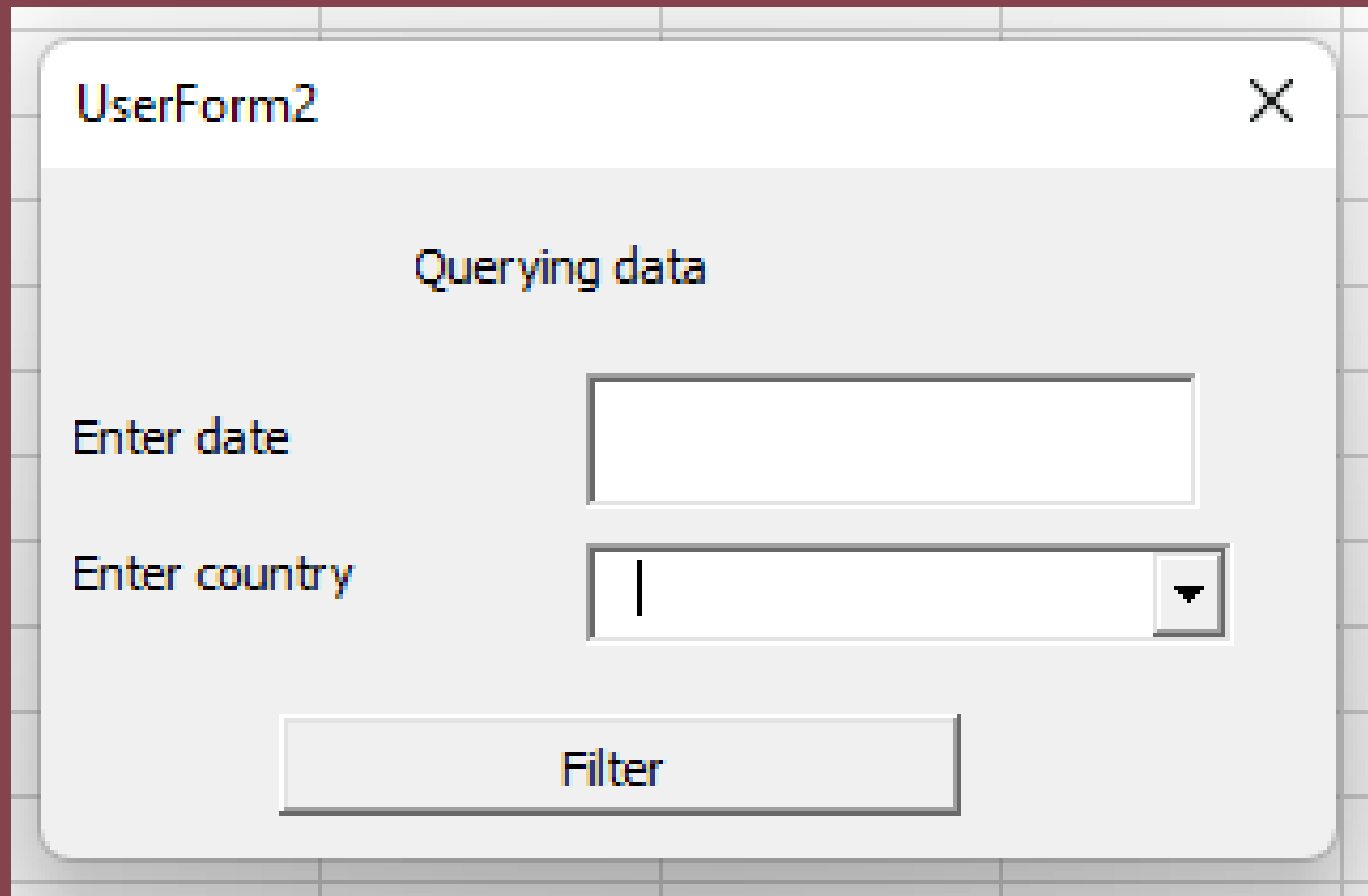
3-b- Forecasting future values

```
Sub Bouton10_Cliquer()  
    Dim ws As Worksheet  
    Dim lastRow As Long  
    Dim nextT As Double  
    Dim nextValue As Double  
  
    Set ws = ThisWorkbook.Worksheets("soil_pollution_diseases")  
  
    lastRow = ws.Cells(ws.Rows.Count, "B").End(xlUp).Row  
  
    nextT = lastRow + 1  
  
    nextValue = WorksheetFunction.Forecast(nextT, ws.Range("F2:F" & lastRow), ws.Range("B2:B" & lastRow))  
  
    MsgBox "Forecast Pollutant Concentration for next time step (" & nextT & ") = " & Format(nextValue, "0.00") & " mg/kg"  
  
End Sub
```



Output and message box.

4- Data querying & Data input



UserForm2

Querying data

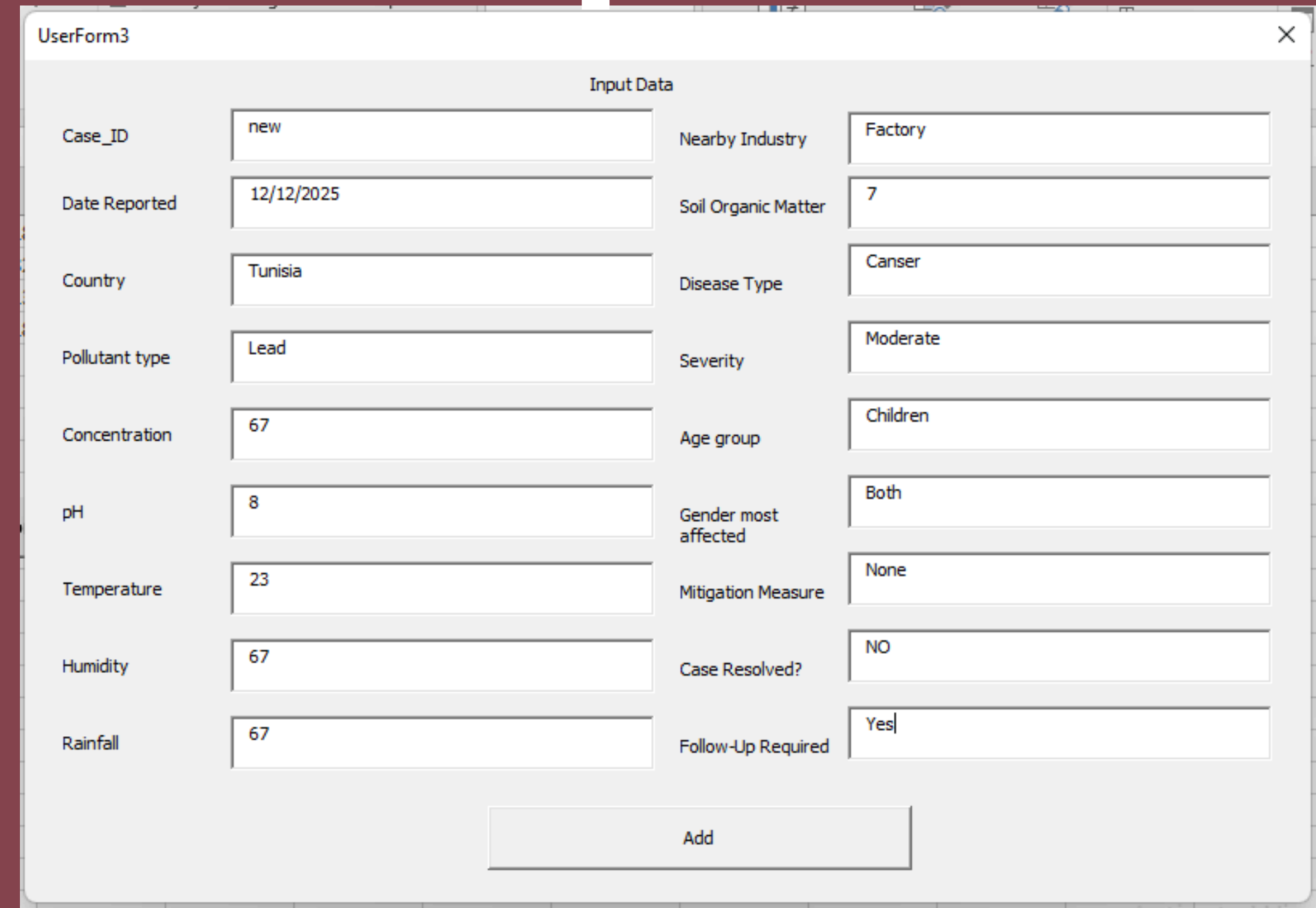
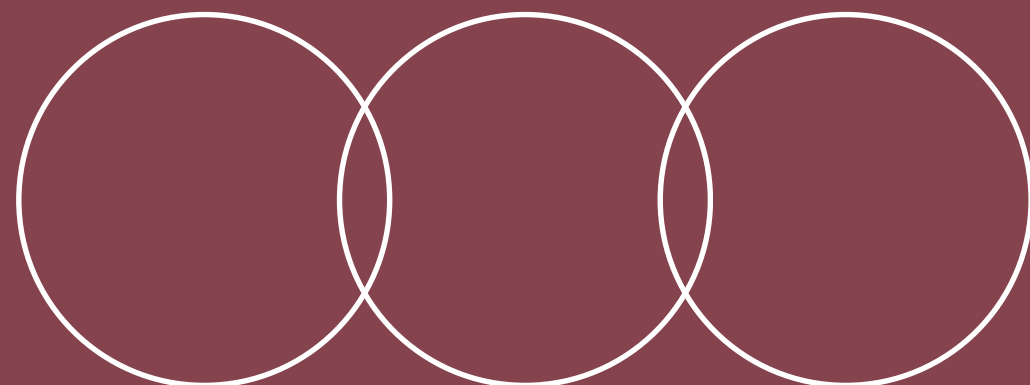
Enter date

Enter country

Filter

This form is used for querying data. It features a title bar 'UserForm2' with a close button. The main area is titled 'Querying data'. It contains two input fields: 'Enter date' and 'Enter country'. The 'Enter country' field has a dropdown arrow on its right side. Below these fields is a 'Filter' button.

Finds pollutant data for a given country and date.



UserForm3

Input Data

Case_ID	new	Nearby Industry	Factory
Date Reported	12/12/2025	Soil Organic Matter	7
Country	Tunisia	Disease Type	Cancer
Pollutant type	Lead	Severity	Moderate
Concentration	67	Age group	Children
pH	8	Gender most affected	Both
Temperature	23	Mitigation Measure	None
Humidity	67	Case Resolved?	NO
Rainfall	67	Follow-Up Required	Yes

Add

This form is used for inputting data. It features a title bar 'UserForm3' with a close button. The main area is titled 'Input Data'. It contains a table with 8 rows and 4 columns. The columns are labeled with various data fields. The table is filled with sample data. Below the table is an 'Add' button.

Adds new data to the set.

5- Highlighting thresholds for ecological risk

```
Sub Bouton2_Cliquer()  
    Dim i As Long  
    Dim last_row As Long  
    Dim n As Long  
    Dim percentage As Double  
    Dim x As Long  
    Dim j As Long  
  
    last_row = Cells(Rows.Count, 1).End(xlUp).Row  
    n = 0  
    For i = 2 To last_row  
        If Cells(i, "E").Value > 151.5 Then  
            n = n + 1  
            Cells(i, "E").Interior.Color = vbRed  
        End If  
        For j = 1 To 20  
            If Cells(1, j).Value = "Disease_Severity" Then  
                x = j  
            End If  
        Next j  
        If Cells(i, x).Value = "Mild" Then  
            Cells(i, x).Interior.Color = vbGreen  
        ElseIf Cells(i, x).Value = "Moderate" Then  
            Cells(i, x).Interior.Color = vbYellow  
        Else  
            Cells(i, x).Interior.Color = vbRed  
        End If  
    Next i  
    total_rows = last_row - 1  
    percentage = (n / total_rows) * 100  
    MsgBox "The upper quartile of Pollutant Concentration is  
  
End Sub
```

Alignement				Nombre		Style			Cellules	
E	F	G	H	I	J	K	L	M	N	
121.59	8.19	13.1	34.7	18.1	Textile	Rainwater	3.59	Cancer	Moderate	Child
21.28	8.03	38.1	57.9	386.9	Chemical	Lake	4.47	Neurological	Severe	Elder
146.98	6.9	35.6	77.7	390.2	None	Irrigation Can	5.95	Neurological	Mild	Child
32.79	6.5	31.6	72.7	223.9	Mining	Rainwater	0.59	Neurological	Mild	Elder
178.68	7.37	12.6	81.6	197.2	None	Rainwater	8.54	Gastrointest	Severe	Elder
41.35	4.98						3.82	Gastrointest	Severe	Child
191.33	4.57						9.03	Gastrointest	Moderate	Adult
42.6	5.26						2.46	Skin Disease	Moderate	Child
163.18	4.63						0.97	Cancer	Moderate	Adult
43.82	6.85						2.84	Gastrointest	Severe	Child
140.13	4.52						2.16	Neurological	Severe	Child
88.18	5.24						8.29	Respiratory I	Moderate	Elder

Microsoft Excel

The upper quartile of Pollutant Concentration is about 151.5 mg/kg.
749 cases (24.96666666666667% of the dataset) are above this level and
will be flagged as HIGH RISK (highlighted in red)

OK

Output and message box.

Thank you.

